# Taylor's and Maclaurin's Series

- $$f(x+a) = f(a) + \frac{x f'(a)}{1!} + \frac{x^2 f''(a)}{2!} + \frac{x^3 f'''(a)}{3!} + \frac{x^4 f^{iv}(a)}{4!} + \cdots + \underbrace{\frac{a^n}{n!} f^n(x+\theta a)}_{\text{error or remainder form.}}$$

Taylor's Series

$f(x+a) \rightarrow$ transforms in $f(x)$ form.

Maclaurin's series :-

- Same formula, take $a=0$.

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots + \underbrace{\frac{x^n}{n!} f^n(\theta x)}_{\text{error or remainder form.}}$$

In question, ask about to find Sin 32, cos 91 etc.

$$32^\circ = \frac{30}{x} + \frac{2}{a}$$ Wrong way ✗

write vice versa

$$32^\circ = \frac{2}{a} + \frac{30^\circ}{x}$$

Eg: Sin 32°

$$f(x+a) = f(x)$$

$$f\left(\frac{2^\circ}{x} + \frac{30^\circ}{a}\right) = f(32^\circ)$$

$$f(x+a) = f(30) + 2^\circ f'(30) + \frac{(2^\circ)^2}{2!} f''(30^\circ) + \cdots$$

Question :-

1.) Find the taylors series upto $4^\circ$ of $f(x) = e^{2x}$ about $x = 0$.

2.) Find the taylor's series upto $3^\circ$ of $f(x) = \ln(1+x)$ about $x = 0$

3.) Prove that :

$$\log(\sin x) = \log(\sin a) + (x-a)\cot a - \frac{1}{2}(x-a)^2 \operatorname{cosec}^2 a$$

4.) Find the taylor's series upto $3^\circ$ of $f(x) = \log \cos x$ at point $x = \frac{\pi}{3}$

5.) Expand $\sin x$ in the power of $(x - \frac{\pi}{2})$ at ~~point x~~ $4^\circ$.

6.) Expand $\tan^{-1} x$ in the power of $(x-1)$ upto 4 terms.

7.) Expand $\log_e x$ in the power of $(x-1)$ upto four terms. Hence find the value of $\log(1.1)$.

8) Expand taylor's series for the polynomial $2x^3 + 7x^2 + x - 6$ in power of $x - 1$.

9.) find $\cos 32^\circ$ upto $4^\circ$.

10.) Using maclaurin series Expand $\log(1+x)$. Hence reduced to $\log\left(\frac{\sqrt{1+x}}{\sqrt{1-x}}\right) =$

$$x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots$$

11.) Expand using Maclaurin's series $f(x) = e^{x \sin x}$ upto 2 terms

12.) Prove $f(x) = \log(1+\sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \cdots$

13) Prove:- $\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} + \dots$

14.) Find Maclaurin's series of

$f(x) = e^x \cdot \cos x$ upto 3 terms.

## Question

1. Find the taylor's series upto 4° of $f(x) = e^{2x}$ about $x = 0$.

$$f(x+a) = f(\underset{x}{x} + \underset{a}{0}) = f(x)$$

$$a = 0$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0)$$

$$f(a) = e^{2a}$$

$$f(0) = e^{2\cdot 0} = e^0 = 1$$

$$f'(a) = 2e^{2a}$$

$$f'(0) = 2\cdot e^{2\cdot 0} = 2\cdot 1 = 2$$

$$f''(a) = 4e^{2a}$$

$$f''(0) = 4\cdot e^{2\cdot 0} = 4\cdot 1 = 4$$

$$f'''(a) = 8e^{2a}$$

$$f'''(0) = 8\cdot e^{2\cdot 0} = 8\cdot 1 = 8$$

$$f^{iv}(a) = 16e^{2a}$$

$$= 16\cdot e^{2\cdot 0} = 16\cdot 1 = 16$$

$$f(x) = 1 + x\cdot 2 + \frac{x^2}{2!} 4 + \frac{x^3}{3!}\cdot 8 + \frac{x^4}{4!}\cdot 16$$

$$= 1 + 2x + \frac{x^2}{2}\cdot 4 + \frac{x^3}{6}\cdot 8 + \frac{x^4}{24}\cdot 16$$

$$= 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4$$

2. $f(x) = \ln(1+x)$ about $x = 0$ about 3°

$$f(x) = \ln(1+x)$$

$$f(\underset{x}{x} + \underset{a}{0}) = f(x)$$

$$f(x) = f(0) + x \cdot f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0)$$

$$f(a) = \ln(1+a)$$

$$f(0) = \ln(1+0) = \ln(1) = 0$$

$$f'(a) = \frac{1}{1+a}$$

$$f'(0) = \frac{1}{1+0} = 1$$

$$f''(a) = \frac{-1}{(1+a)^2}$$

$$f''(0) = \frac{-1}{(1+0)^2} = -1$$

$$f'''(a) = \frac{2}{(1+a)^3}$$

$$f'''(0) = \frac{2}{(1+0)^3} = \frac{2}{1} = 2$$

$$f(x) = 0 + x \cdot 1 + \frac{x^2}{2!}(-1) + \frac{x^3}{3!} 2$$

$$= x + \frac{x^2}{2}(-1) + \frac{x^3}{6} \cdot 2$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3}$$

Prove that :

$$\log(\sin x) = \log(\sin a) + (x-a)\cot a - \frac{1}{2}(x-a)^2 \cdot \operatorname{cosec}^2 a$$

$$f(x) = \log(\sin x)$$

$$f(x+a) = f(\underbrace{x-a}_{x} + \underbrace{a}_{a}) = f(x)$$

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a)$$

$$f(a) = \log(\sin a)$$

$$f'(a) = \frac{1}{\sin a} \cdot \cos a = \cot a$$

$$f''(a) = -\operatorname{cosec}^2 a$$

$$f(x) = \log(\sin a) + (x-a)\cdot\cot a + \frac{(x-a)^2}{2!} \cdot (-\operatorname{cosec}^2 a)$$

$$\log(\sin x) = \log \sin a + (x-a)\cot a - \frac{1}{2}(x-a)^2 \cdot \operatorname{cosec}^2 a$$

proved//.

4. Find the taylor's series expansion of $\log(\cos x)$ at point $x = \frac{\pi}{3}$ at 3:

$f(x) = \log(\cos x)$

at point $x = \pi/3$

$$\left(\underline{x - \pi/3}\right) = 0$$

$$f(x+a) = f\left(\underbrace{x - \pi/3}_{x} + \underbrace{\pi/3}_{a}\right) = f(x)$$

$$f(x) = f(a) + x \cdot f'(a) + \frac{x^2}{2!} f''(a) + \frac{x^3}{3!} f'''(a)$$

$$f(x) = f(\pi/3) + (x - \pi/3) f'(\pi/3) + \frac{(x-\pi/3)^2}{2!} f''(\pi/3) + \frac{(x-\pi/3)^3}{3!} f'''(\pi/3)$$

$f(a) = \log(\cos a)$

$f(\pi/3) = \log(\cos \pi/3) = \log(1/2)$

$f'(a) = \frac{1}{\cos a} \cdot (-\sin a) = -\tan a$

$f'(\pi/3) = -\tan \pi/3 = -\sqrt{3}$

$f''(a) = -\sec^2 a$

$f''(\pi/3) = -\sec^2 \pi/3 = -4$

$f'''(a) = -2\sec^2 a \tan a$

$f'''(\pi/3) = -8\sqrt{3}$

$$f(x) = \log(1/2) + (x - \pi/3)\cdot(-\sqrt{3}) + \frac{(x-\pi/3)^2}{2!}\cdot(-4)$$
$$+ \frac{(x-\pi/3)^3}{3!}(-8\sqrt{3})$$

$$\log(\cos x) = \log(1/2) - \sqrt{3}(x - \pi/3) - 2(x - \pi/3)^2$$
$$- \frac{4}{\sqrt{3}}(x - \pi/3)^3$$

5. Expand $\sin x$ in power of $(x - \pi/2)$ at ~~a point~~ ~~$x = \pi/2$~~ at 4.

$f(x) = \sin x$

power of $(x - \pi/2)$

$$f(x+a) = f(\underbrace{x - \pi/2}_{x} + \underbrace{\pi/2}_{a}) = f(x)$$

$f(a) = \sin a$

$f(\pi/2) = \sin \pi/2 = 1$

$f'(a) = \cos a$

$f'(\pi/2) = \cos \pi/2 = 0$

$f''(a) = -\sin a$

$f''(\pi/2) = -\sin \pi/2 = -1$

$f'''(a) = -\cos a$

$f'''(\pi/2) = -\cos \pi/2 = 0$

$f^{iv}(a) = -(-\sin a) = \sin a$

$f^{iv}(\pi/2) = \sin \pi/2 = 1$

$$f(x) = f(\pi/2) + (x - \pi/2) \cdot f'(\pi/2) + \frac{(x-\pi/2)^2}{2!} f''(\pi/2)$$

$$+ \frac{(x-\pi/2)^3}{3!} f'''(\pi/2) + \frac{(x-\pi/2)^4}{4!} f^{iv}(\pi/2)$$

$$f(x) = 1 + (x - \pi/2) \cdot 0 + \frac{(x-\pi/2)^2}{2!} \cdot (-1)$$

$$+ \frac{(x-\pi/2)^3}{3!} \cdot 0 + \frac{(x-\pi/2)^4}{4!} \cdot (1)$$

$$f(x) = 1 - \frac{1}{2}(x - \pi/2)^2 + \frac{(x - \frac{\pi}{2})^4}{24}$$

$$\sin x = 1 - \frac{1}{2}(x - \pi/2)^2 + \frac{1}{24}(x - \pi/2)^4$$

$f(x) = \tan^{-1}(x)$ in power of $(x-1)$ upto 4 terms.

$f(x) = \tan^{-1}(x)$

power of $(x-1)$

$f(x+a) = f\left(\underbrace{x-1}_{x} + \underbrace{1}_{a}\right) = f(x)$

$$f(x) = f(a) + f'(a)\cdot x + \frac{x^2}{2!} f''(a) + \frac{x^3}{3!} \cdot f'''(a) + \frac{x^4}{4!} f^{iv}(a)$$

$$f(x) = f(1) + f'(1)\cdot(x-1) + \frac{(x-1)^2}{2!} f''(1) + \frac{(x-1)^3}{3!} f'''(1) + \frac{(x-1)^4}{4!} f^{iv}(1)$$

$f(a) = \tan^{-1}(a)$

$f(1) = \tan^{-1}(1) = \pi/4$

$f'(a) = \dfrac{1}{1+a^2}$

$f'(1) = \dfrac{1}{1+1^2} = \dfrac{1}{2}$

$f''(a) = -\dfrac{1}{(1+a^2)^2} \cdot 2a$

$$f''(1) = \frac{-1}{(1+1^2)^2} \cdot 2 \cdot 1$$

$$= -\frac{1}{2}$$

$$f'''(a) = -\left[ +\frac{2a}{(1+a^2)^2} + 2a \cdot \frac{(-2a) \cdot 2}{(1+a^2)^3} \right]$$

$$= -\left[ \frac{2}{(1+a^2)^2} - \frac{4a^2 \times 2}{(1+a^2)^3} \right]$$

$$f'''(1) = -\left[ \frac{2}{(1+1^2)^2} - \frac{4 \cdot 1^2 \times 2}{(1+1^2)^3} \right]$$

$$= -\left[ \frac{2}{4} - \frac{4 \times 2}{8} \right]$$

$$= -\left[ \frac{2}{4} - \frac{2 \times 2}{4} \right]$$

$$= +\frac{1 \times 2}{4} = \frac{1}{2}$$

$$f^{iv}(a) = +\frac{8 \cdot a}{(1+a^2)^3} - \frac{48a^3}{(1+a^2)^4} + \frac{16a}{(1+a^2)^3}$$

$$= \frac{8}{2^3} - \frac{48}{2^4} + \frac{16}{2^3}$$

$$= 1 - 3 + 2 = 0$$

$$f(x) = \pi/4 + (x-1)\cdot\frac{1}{2} + \frac{(x-1)^2}{2!}\left(-\frac{1}{2}\right) + \frac{(x-1)^3}{3!}\cdot\frac{1}{2}$$

$$+ \frac{(x-1)^4}{4!} \times 0$$

$$\tan^{-1}x = \pi/2 + \frac{(x-1)}{2} - \frac{(x-1)^2}{4} + \frac{(x-1)^3}{12}$$

Q7. Expand $\log_e x$ in power of $(x-1)$ upto four terms. Hence find the value of $\log(1.1)$.

$f(x) = \log_e x$

power of $(x-1)$

$$f(x+a) = f\left(\underbrace{x-1}_{x} + \underbrace{1}_{a}\right) = f(x)$$

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1)$$

$$+ \frac{(x-1)^4}{4!}f^{iv}(1)$$

$f(a) = \log a$

$f(1) = \log 1 = 0$

$f'(a) = \frac{1}{a}$

$f'(1) = 1$

$$f''(a) = \frac{-1}{a^2}$$

$$f''(1) = -1$$

$$f'''(a) = +\frac{2}{a^3}$$

$$f'''(1) = 2$$

$$f^{iv}(a) = -\frac{6}{a^4}$$

$$f^{iv}(1) = -6$$

$$f(x) = 0 + (x-1)\cdot 1 + \frac{(x-1)^2}{2!}\cdot(-1) + \frac{(x-1)^3}{3!}(2) + \frac{(x-1)^4}{4!}\cdot(-6)$$

$$\log x = (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4}$$

Now to find $\log(1.1)$

at $x = 1.1$

$$\log(1.1) = (1.1-1) - \frac{(1.1-1)^2}{2} + \frac{(1.1-1)^3}{3} - \frac{(1.1-1)^4}{4}$$

$\log(1.1) = 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} + \frac{(0.1)^4}{4}$

8) Expand taylor's seriers for the polynomial $2x^3 + 7x^2 + x - 6$ in power of $x-1$.

$f(x) = 2x^3 + 7x^2 + x - 6$

power of $x-1$

$f(x+a) = f(\underbrace{x-1}_{x} + \underbrace{1}_{a}) = f(x)$

$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \frac{(x-1)^3}{3!}f'''(1)$
$+ \frac{(x-1)^4}{4!}f^{iv}(1)$

$f(a) = 2a^3 + 7a^2 + a - 6$

$f(1) = 2.1^3 + 7.1^2 + 1 - 6$
$= 4$

$f'(a) = 6a^2 + 14a + 1$

$f'(1) = 6.1^2 + 14.1 + 1$
$= 21$

$f''(a) = 12a + 14$

$f''(1) = 12.1 + 14$
$= 26$

$f''(a) = 12$

$f'''(1) = 12$

$f^{iv}(a) = 0$

$f^{iv}(1) = 0$

$$f(x) = 4 + (x-1)\cdot 21 + \frac{(x-1)^2}{2!}\cdot 26$$

$$+ \frac{(x-1)^3}{3!}\cdot 12 + \frac{(x-1)^4}{4!}\cdot 0$$

$$f(x) = 4 + 21(x-1) + 13(x-1)^2 + 2(x-1)^3$$

Q.) find $(\cos(32^\circ))$ upto 4

$$f(x) = \cos(32^\circ)$$

$$f(x+a) = f(\underbrace{2^\circ}_{x} + \underbrace{30^\circ}_{a}) = f(32^\circ) = f(x)$$

$$f(x) = f(30^\circ) + 2^\circ f'(30^\circ) + \frac{(2^\circ)^2}{2!} f''(30^\circ)$$

$$+ \frac{(2^\circ)^3}{3!} f'''(30^\circ) + \frac{(2^\circ)^4}{4!} f^{iv}(30^\circ)$$

$$f(x) = f(\pi/6) + \left(\frac{\pi}{90}\right) f'(\pi/6) + \frac{\left(\frac{\pi}{90}\right)^2}{2!} f''(\pi/6)$$

$$+ \frac{(\pi/90)^3}{3!} f'''(\pi/6) + \frac{(\pi/90)^4}{4!} f^{iv}(\pi/6)$$

$f(a) = \cos a$

$f(\pi/6) = \cos(\pi/6) = \frac{\sqrt{3}}{2}$

$f'(a) = -\sin a$

$f'(\pi/6) = -\sin(\pi/6) = -\frac{1}{2}$

$f''(a) = -\cos a$

$f''(\pi/6) = -\cos(\pi/6) = -\sqrt{3}/2$

$f'''(a) = \sin a$

$f'''(\pi/6) = \sin(\pi/6) = \frac{1}{2}$

$f^{iv}(a) = \cos a$

$f^{iv}(\pi/6) = \cos(\pi/6) = \frac{\sqrt{3}}{2}$

$$f(x) = \frac{\sqrt{3}}{2} + \frac{\pi}{90} \cdot \left(-\frac{1}{2}\right) + \frac{\pi}{90 \cdot (2!)} \left(-\frac{\sqrt{3}}{2}\right) + \frac{\pi}{90 \cdot (3!)} \cdot \frac{1}{2}$$

$$+ \frac{\pi}{90 \cdot (4!)} \cdot \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{2} + \frac{3.14}{90}\left(-\frac{1}{2}\right) + \frac{3.14}{90 \cdot 2}\left(-\frac{\sqrt{3}}{2}\right) + \frac{3.14}{90 \cdot (6)} \times \frac{1}{2}$$

$$= \frac{3.14}{90 \cdot (24)} \cdot \frac{\sqrt{3}}{2}$$

Q.) Using Maclaurin's series Expand $\log(1+x)$
Hence reduced to $\log\left(\sqrt{\frac{1+x}{1-x}}\right) =$

$$x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots$$

$$f(x) = \log(1+x)$$

$$f(x) = f(0) + x \cdot f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0)$$

$$+ \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^{v}(0) + \cdots$$

$$f(x) = \log(1+x)$$

$$f(0) = \log(1+0) = \log(1) = 0$$

$$f'(x) = \frac{1}{1+x}$$

$$f'(0) = 1$$

$$f''(x) = -\frac{1}{(1+x)^2}$$

$$f''(0) = -1$$

$$f'''(x) = \frac{2}{(1+x)^3}$$

$$f'''(0) = 2$$

$$f^{iv}(a) = \frac{-6}{(1+x)^4}$$

$$f^{iv}(0) = -6$$

$$f^{v}(a) = \frac{24}{(1+x)^5}$$

$$f^{v}(0) = 24$$

$$f(x) = 0 + x \cdot (1) + \frac{x^2}{2!} \cdot (-1) + \frac{x^3}{3!}(2)$$

$$+ \frac{x^4}{4!}(-6) + \frac{x^5}{5!} \cdot 24 + \cdots$$

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5}$$

Now $\log\sqrt{\frac{1+x}{1-x}}$

$$\log\left(\frac{1+x}{1-x}\right)^{\frac{1}{2}}$$

$$\frac{1}{2} \cdot \log\left(\frac{1+x}{1-x}\right)$$

$$\frac{1}{2}\left[\log(1+x) - \log(1-x)\right]$$

$g(x) = \log(1-x)$

$g(a) = \log(1-a)$

$g(0) = \log(1) = 0$

$g'(a) = \frac{1}{1-a}(-1)$

$g'(0) = -1$

$g''(a) = \frac{1}{(1-a)^2}(-1)$

$g''(0) = -1$

$g'''(a) = \frac{+2}{(1-a)^3}(-1)$

$g'''(0) = -2$

$g^{iv}(a) = \frac{-6}{(1-a)^4}$

$g^{iv}(0) = -6$

$g^{v}(a) = \frac{-24}{(1-a)^5}$

$g^{v}(0) = -24$

$$g(x) = 0 + x \cdot (-1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!} \cdot -2 + \frac{x^4}{4!} \cdot -6$$

$$\frac{x^5}{5!} \cdot -24$$

$$= -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} + \dots$$

$f(x) - g(x)$

$$\left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{5} + \frac{x^5}{5} + \dots\right] - \left[-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5}\right.$$

$$2x + \frac{2x^3}{2} + \frac{2x^5}{5}$$

$$2 \cdot \left(x + \frac{x^3}{3} + \frac{x^5}{5}\right)$$

$$(\log(1+x) - \log(1-x)) = 2 \cdot \left(x + \frac{x^3}{3} + \frac{x^5}{5}\right)$$

$$\log\left(\frac{1+x}{1-x}\right) = 2 \cdot x + \frac{x^3}{3} + \frac{x^5}{5}$$

$$\frac{1}{2} \log \frac{(1+x)}{(1-x)} = x + \frac{x^3}{3} + \frac{x^5}{5}$$

$$\boxed{\log\left(\sqrt{\frac{1+x}{1-x}}\right) = x + \frac{x^3}{3} + \frac{x^5}{5}}$$

Q.) $e^{x \sin x}$ upto 2 term.

$f(x) = e^{x \cdot \sin x}$

$f(x) = f(0) + x \cdot f'(0) + \frac{x^2}{2!} f''(0)$

$f(a) = e^{a \sin a}$

$f(0) = 1$

$f'(a) = e^{a \sin a} \cdot [a \cos a + \sin a]$

$f'(0) = 1 \cdot [0]$

$= 0$

$f''(a) = e^{a \sin a} [\cos a \cdot a + \sin a] + e^{a \sin a} \cdot [\cos a + a(-\sin a) + \cos a]$

$f''(0) = e^0 [1 \cdot 0 + 0] \cdot e^0 [1 + 0 + 1]$

$= 2$

$f(x) = 1 + x \cdot 0 + \frac{x^2}{2!} + 2$

$f(x) = 1 + x$

Prove

(3) $f(x) = \log(1+\sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12} + \cdots$

$$f(x) = f(0) + x \cdot f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) \cdots$$

$$f(a) = \log(1+\sin a)$$

$$f(0) = 0$$

$$f'(a) = \frac{\cos a}{1+\sin a}$$

$$f'(0) = 1$$

$$f''(a) = \frac{(1+\sin a)(-\sin a) - \cos a \cdot \cos a}{(1+\sin a)^2}$$

$$= \frac{-\sin a - \sin^2 a - \cos^2 a}{(1+\sin a)^2}$$

$$= \frac{-\sin a - 1}{(1+\sin a)^2}$$

$$f''(a) = -\frac{1}{1+\sin a}$$

$$f''(0) = -1$$

$$f'''(a) = (1+\sin a)^{-2} \cdot \cos a$$

$$f'''(0) = 1$$

$$= \frac{\sin a}{(1+\sin a)^2} + \cos a \cdot \frac{(-2)}{(1+\sin a)^3} \cos a$$

$$f^{iv}(a) =$$

$$f^{iv}(0) = -2$$

$$f(x) = 0 + x \cdot (1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!} \cdot 1 + \frac{x^4}{4!}(-2)$$

$$\log(1+\sin x) = x - \frac{x^2}{2} + \frac{x^3}{6} - \frac{x^4}{12}$$

proved

Q.) prove:- $\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} + \dots$

$$f(x) = \log(1+e^x)$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots$$

$$f(a) = \log(1+e^a)$$

$$f(0) = \log 2$$

$$f'(a) = \frac{e^a}{1+e^a}$$

$$f'(0) = \frac{1}{2}$$

$$f''(a) = \frac{e^a(1+e^a) - e^a \cdot e^a}{(1+e^a)^2}$$

$f''(0) = \frac{1}{4}$

$f(x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} + \cdots$

Q. Find Maclaurins Series:-

$f(x) = e^x \cdot \cos x$ upto 3

$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f''(0)$

$f(x) = e^x \cos x$

$f(0) = 1$

$f'(x) = e^x(-\sin x) + \cos x \cdot e^x$

$f'(0) = 1$

$f''(x) = -[\sin x \cdot e^x + e^x \cos x] + [e^x(-\sin x) + \cos x \cdot e^x]$

$f''(0) = -1 + 1 = 0$

~~$f'''(x) = -[\sin x e^x + e^x \cos x] + e^x(-\sin x) + \cos x$~~

$f'''(x) = -2[\sin x \cdot e^x + e^x \cos x]$

$f'''(0) = -2$

$f(x) = 1 + x \cdot 1 + \frac{x^2}{2!} \cdot 0 + \frac{x^3}{3!}(-2)$

$e^x \cos x = 1 + x + \frac{x^2}{2} - \frac{x^3}{3}$